

TANMAY WARKE

AI/ML Engineer | Data Analytics & Python Developer | Full-Stack Developer

Pune, Maharashtra, India | +91 7888249506 | tanmaywarke1227@gmail.com

[Linkedin](#) | [GitHub](#)

PROFILE

Third-year B.Tech Computer Science Engineering student specializing in AI & Analytics, with hands-on experience in Python, machine learning, data analysis, AI model testing, software validation, and technical documentation. Strong problem-solving ability, attention to detail, and practical exposure to real-world software and AI systems. Capable of identifying issues, improving system reliability, and supporting end-to-end technical workflows with a structured, analytical, and execution-focused approach suited to fast-paced engineering environments.

EXPERIENCE

Software Engineer Intern, Uchit Technology Pvt. Ltd. 2026

August 2026 - November

- Contributed to AI-based application development, working with Ultralytics YOLO family models for real-time object detection within an AI-powered video analytics and management system.
- Performed software testing across backend and frontend components, verifying functional accuracy and consistent behavior across the application.
- Conducted system validation and model evaluation, assessing detection accuracy and performance against expected outcomes to ensure reliability before deployment.
- Carried out bug identification and performance analysis to improve system stability and responsiveness, while supporting technical research and production-level architecture planning for real-time detection use cases.

Project-Based Intern, IEEE TechForGood

June 2026

IEEE Student Branch (MIT ADT University), IEEE Region 10 AIPSCC | RL-Based Smart Building Energy Optimization

- Profiled and validated a simulated smart-building dataset of ~43,200 records for occupancy, environmental, and energy trends, ensuring data quality ahead of RL training and evaluation.
- Programmed a Raspberry Pi Pico W (MicroPython) with sensors; caught and fixed critical integration bugs (incorrect burden resistor value, missing RMS conversion) before they reached the data pipeline.
- Contributed to technical proposal documentation and failure-mode analysis; ran end-to-end tests across frontend, backend, RL model, database, and hardware, achieving a 96% pass rate.
- Implemented TD3 reinforcement learning workflows using Stable-Baselines3 and PyTorch, training the agent for 100,000 episodes to optimize HVAC energy usage.
- Built a React.js and TypeScript dashboard integrated with backend REST APIs and PostgreSQL to visualize HVAC decisions and RL outputs.

PROJECTS

Rainfall Prediction Model | Python, Pandas, Scikit-learn, Matplotlib, Seaborn

- Cleaned, explored, and engineered features from raw weather data using Pandas, then built Scikit-learn regression and classification models to predict rainfall, with results visualized in Matplotlib and Seaborn.

AI Character Detector | PyTorch, ResNet50, FastAPI, Grad-CAM, Docker

- Fine-tuned ResNet50 on a labeled dataset of 30,000 images to classify real versus AI-generated content, reaching 99.94% validation accuracy; used Grad-CAM to validate model predictions and built a FastAPI backend with Docker containerization.

EDUCATION

B.Tech in Computer Science Engineering (AI & Analytics)

2024 - 2028

MIT ADT University, Pune, India | CGPA: 8.07

CERTIFICATIONS

- Machine Learning with Python - IBM ([View Certificate](#))
- Introduction to AI - IBM ([View Certificate](#))
- Computer Networking - Google ([View Certificate](#))

LANGUAGES

English (Fluent), Hindi

TECHNICAL SKILLS

Data Processing & Analytics: Python, Pandas, SQL, Scikit-learn, Matplotlib, Seaborn

AI / ML: Deep Learning (PyTorch), Reinforcement Learning (TD3), Stable-Baselines3, Gymnasium

Web & Full-Stack: React.js, TypeScript, JavaScript, Tailwind CSS, Chart.js, Flask, FastAPI, Django, REST APIs

Databases & Tools: PostgreSQL, Git, GitHub, Docker

IoT & Embedded: Raspberry Pi Pico W (MicroPython), IoT Sensor Integration